

Agenda:

1. Single Number
2. Find Two Missing Numbers
3. Sum of Xor of all Pairs

$$A \oplus 0 = A \quad A \oplus A = 0$$

$$5 \oplus 7 \oplus 5 \rightarrow (5 \oplus 5) \oplus 7 = 0 \oplus 7 = \underline{7}$$

Q1: Given an integer array **A[]** where every number occurs **twice** except for one number which occurs just **once**. Find that number.

Example: $A = [4, 5, 5, 4, 1, 6, 6]$ $\text{ans} = 1$

$A = [7, 5, 5, 1, 7, 6, 1, 6, 4]$ $\text{ans} = 4$

$$4 \oplus 5 \oplus 5 \oplus 4 \oplus 1 \oplus 6 \oplus 6 \rightarrow (4 \oplus 4) \oplus (5 \oplus 5) \oplus (6 \oplus 6) \oplus 1$$

$$= (0 \oplus 0) \oplus (0 \oplus 0)$$

$$\text{ans} = \forall i A[i]$$

$$= 0 \oplus 1 = \underline{1}$$

	3	2	1	0		0	1	2	3	4	5	6		
					A[]	→	[	5	5	4	11	11	9	4]
5 →	0	1	0	1										
5 →	0	1	0	1										
4 →	0	1	0	0										
11 →	1	0	1	1										
11 →	1	0	1	1										
9 →	1	0	0	1										
4 →	0	1	0	0										

odd 1's at any position ⇒
unique number has that bit = 1

$\wedge$ for even 1's = 0
 $\wedge$ or odd 1's = 1

2 1 0

odd 1's at any position ⇒
unique number has that bit = 1

^ for even 1's = 0
^ or odd 1's = 1

#1's → 3 4 2 5
1 0 0 1 → 9

All numbers occur 3 times
except 1 unique number.

	2	1	0
5 →	1	0	1
6 →	1	1	0
5 →	1	0	1
4 →	1	0	0
6 →	1	1	0
5 →	1	0	1
6 →	1	1	0
#1's →	<u>7</u>	<u>3</u>	<u>3</u>
	1	0	0 → <u><u>4</u></u>

Q2: Given an array **A[]** of length N where all the elements are distinct and are in the range **[1, N+2]**. Two numbers from the range **[1, N+2]** are missing from the array A. Find the two missing numbers.

Example: A = [5, 1, 3, 6] Range: [1 6] ans = {2, 4}

A = [3, 2, 4] Range: [1 5] ans = {1, 5}

Sol 1 → Sort A[]

	i	i	i	i	i
	0	1	2	3	
A →	[1	3	5	6]	

cnt = 1 2 3 4 5 6 7

cnt = 1 ans → 2, 4

for i → 0 to (N-1) {

if (A[i] != cnt) {

print (cnt)

i--

}

cnt++

}

TC = $O(N \log(N))$

$O(N)$

Sol 2 → [1 (N+2)] A[] → all but 2 numbers

let say x, y

$$\text{Sum} \rightarrow 1 \text{ to } (N+2) = \frac{(N+2) * (N+3)}{2} = S$$

$$\sum A[i] + x + y = S \Rightarrow x + y = S - \sum A[i] \quad (i)$$

$$\text{Square Sum} \rightarrow 1^2 + 2^2 + \dots + (N+2)^2 = \frac{(N+2) * (N+3) * (2(N+2) + 1)}{6} = T$$

$$\sum (A[i])^2 + x^2 + y^2 = T \Rightarrow x^2 + y^2 = T - \sum (A[i])^2 \quad (ii)$$

$$TC = O(N) \quad x^2 + (S - \sum A[i] - x)^2 = T - \sum (A[i])^2$$

Sol 3 $\rightarrow$ $A = [5 \ 1 \ 3 \ 6] \quad [1 \ 6]$

5 1 3 6 1 2 3 4 5 6

Steps $\rightarrow 1 \rightarrow \forall i \wedge A[i] \wedge [1 - (N+2)] \quad x \wedge y = K$

$$x \neq y \Rightarrow \underline{x \wedge y > 0}$$

$\Rightarrow$ K has atleast 1 bit set

$$x \wedge y = 2^4 = 6 \rightarrow \begin{matrix} 2 & 1 & 0 \\ 1 & 1 & 0 \end{matrix}$$

$\swarrow \searrow$

$\begin{matrix} 0 & 1 & 0 \\ 1 & 0 & 0 \end{matrix}$

$\rightarrow \text{pos} = 1$

2 $\rightarrow$ Find a pos where $x \wedge y$ has 1

$\Rightarrow$ either x has that bit = 1 or y has that bit = 1

pos = -1

// xor = $x \wedge y$

for $i \rightarrow 0$ to 31 {

if $((xor \& (1 \ll i)) > 0)$ {

pos = i

break

}

}

1 $\rightarrow$ 001

5 $\rightarrow$ 101

2 $\rightarrow$ 010

6 $\rightarrow$ 110

3 $\rightarrow$ 011

4 $\rightarrow$ 100

3 $\rightarrow$

5

1

3

6

1

2

3

4

5

6

pos = 1

0
1

5 1 1 4 5 $\rightarrow$ xor all = 4

3 6 2 3 6 $\rightarrow$ xor all = 2

x = 0 y = 0

for $i \rightarrow 0$ to (N-1) {

if $((A[i] \& (1 \ll pos)) == 0)$ $x \wedge = A[i]$

else $y \wedge = A[i]$ // $y = y \wedge A[i]$

}

for $i \rightarrow 1$ to (N+2) {

if $((i \& (1 \ll pos)) == 0)$ $x \wedge = i$

else $y \wedge = i$ // $y = y \wedge A[i]$

}

return {x, y}

TC = $O(N)$ SC = $O(1)$

Q3: Given an array **A[]** of **N** integers. Find the **sum of bitwise XOR all pairs** of numbers in the array. Since the answer can be large, return the answer % (10^9 + 7).

Example:	A = [3, 4, 2]	Pair	Xor
		{3, 4}	7
		{3, 2}	1
		{4, 2}	6
		Sum of xor of all pairs = 7 + 1 + 6 = 14.	

0 1 2 3 4 5

A = [2, 3, 5, 7, 11, 13]

2³ 2² 2¹ 2⁰

3 2 1 0

2 → 0 0 1 0

3 → 0 0 1 1

5 → 0 1 0 1

7 → 0 1 1 1

11 → 1 0 1 1

13 → 1 1 0 1

#0's → 4 3 2 1

#1's → 2 3 4 5

8 9 8 5

contri → 8*2³ = 64

9*2² = 36

5*2⁰ = 5

8*2 = 16

B₁ Box 1

B₂ Box 2

B₃

3 * 2 = 6

ans = 64 + 36 + 16 + 5

= 121

ans = 0

mod = $10^9 + 7$

for b → 0 to 31 {

x = 0

y = 0

// x → # 0's y → # 1's

for i → 0 to (N-1) {

if ((A[i] & (1 << i)) == 0) x++

else y++

}

ans += (x * y) % mod * (1 << b) % mod

ans = ans % mod

}

TC = $O(N)$

SC = $O(1)$

return ans